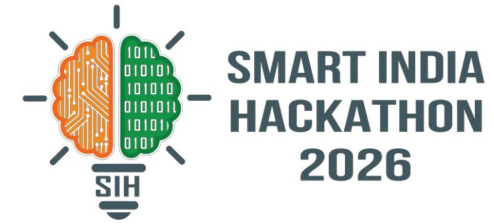
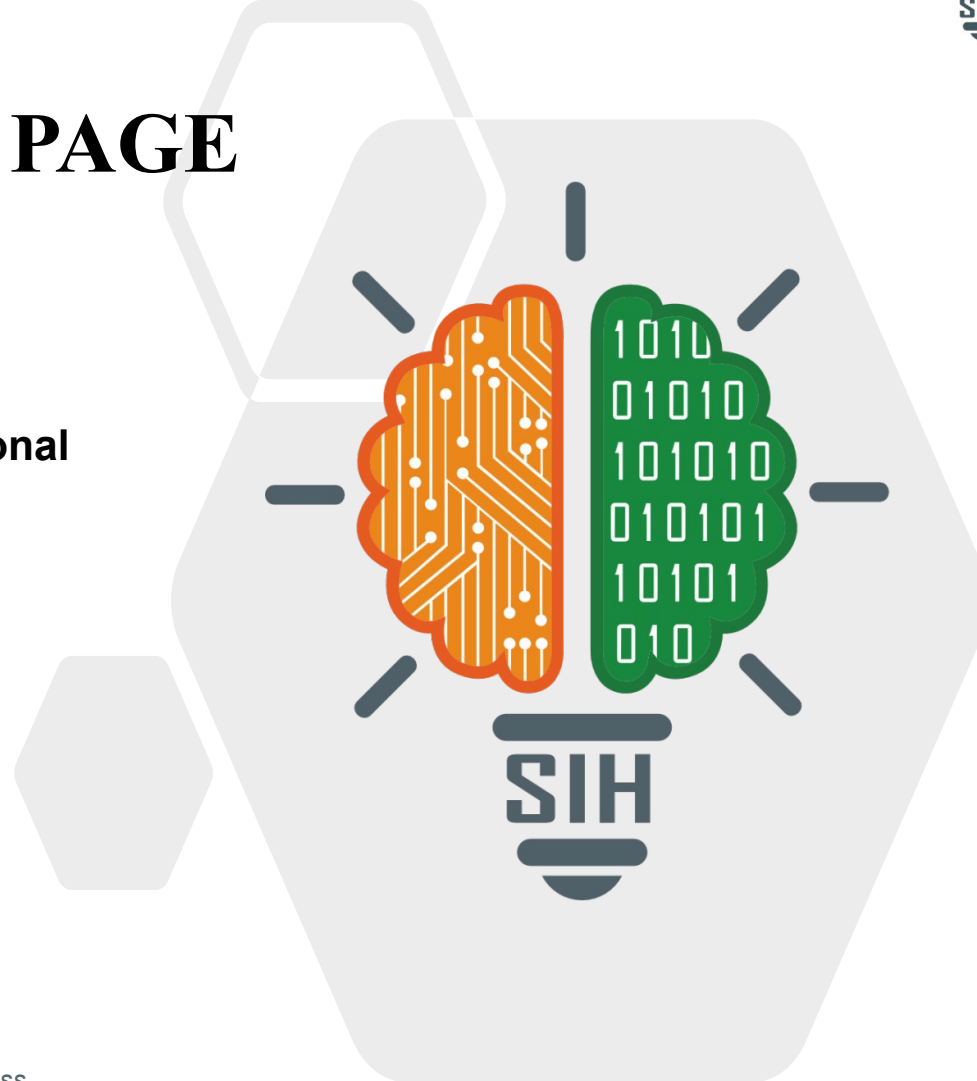


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – SIH26068
- Problem Statement Title- WeatherGPT: Conversational AI for Weather Forecasting, Alerts, and Climate Information
- Theme- Disaster Management
- PS Category- Software
- Team ID- 118709
- Team Name- Void Pointers



WeatherGPT: ask the sky, trust the answer.

Source-linked weather conversations, with Indian-language and voice access.



IMD's forecasts, in your language. Every number comes with a receipt.

How it addresses the problem

✗ Weather data is scattered across portals, bulletins, satellite products & forecast systems

➔

One conversational interface: actionable insight in seconds

✗ Generic AI chatbots invent forecasts, even warnings

➔

Grounded: values checked against source records

✗ English-only, text-heavy portals shut out rural India

➔

Indian-language answers & voice input; acceptance ongoing

WeatherGPT

ગુજરાતી

Surat, Gujarat

કાલે સવારે સુરતમાં વરસાદ પડશે?

FORECAST RAINFALL

0.4 mm

સુરત • કાલે 09:30–12:30

09:30 evidence covers window 12:30

IMD • Green: no warning

PRAMAAN • GFS 12Z • f+18–21 h
cell 21.25N 72.75E • 23:03 IST

સાંજે? કોઈ ચેતવણી?

તમારો પ્રશ્ન પૂછો...

Concept screen · Gujarati in, Gujarati out



💡 Proposed Solution: seven modules, one conversation

Mausam Setu
IMD-first real-time data

Satya Gate
evidence-bound value checks

Pramaan Receipt
proof on every answer

Vaani
Selected languages,
voice input

Chetavani Relay
early warning dissemination

Krishi Salah
location-based advisory

Jalvayu Lens
climate trend & historical analysis

PS SIH26068 · ALL 8 FEATURES ADDRESSED

✓ Real-time

✓ NL querying

✓ GFS / IMD

✓ Early warning

✓ Location

✓ Indian languages

✓ Climate trend

✓ Voice

INNOVATION & UNIQUENESS

🛡️ Checks before display: tool-owned values; disclosed fallback.

📶 IMD first: IMD's warnings, nowcasts & Mausamgram, beside global models.

🏠 Deployment goal: Indian hosting; configured providers used now.

WATCH DEMO

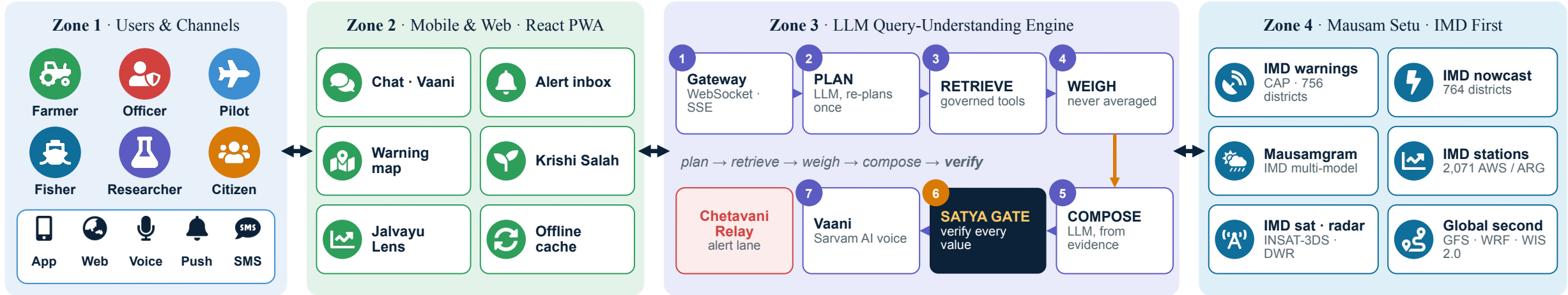
Scan or click QR

@SIH Idea Submission · SIH26068 · WeatherGPT · Team Void Pointers (118709)

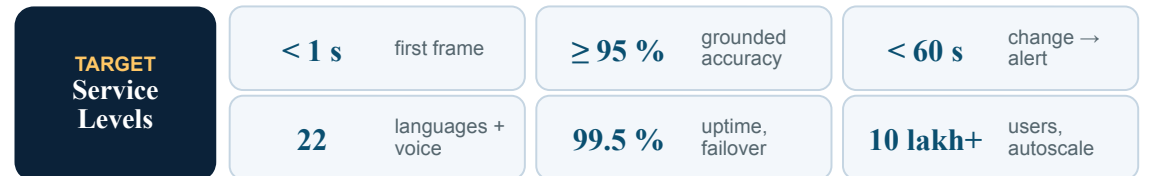
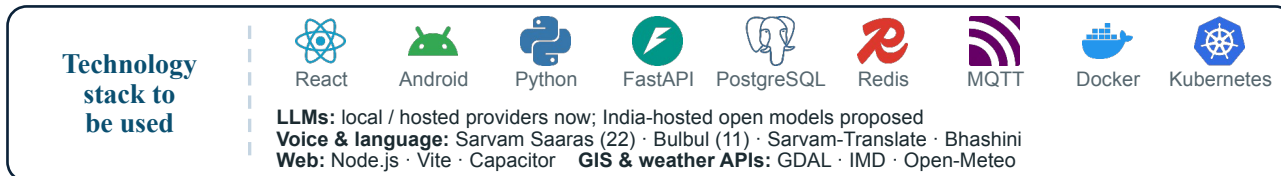
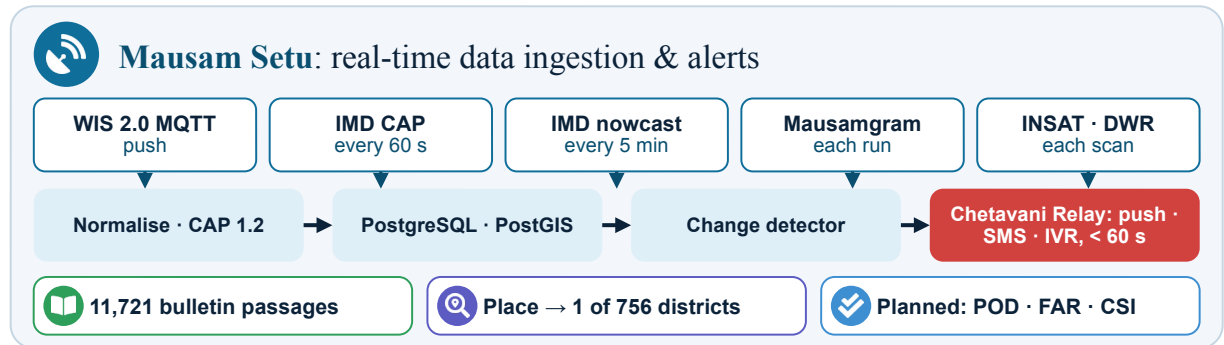
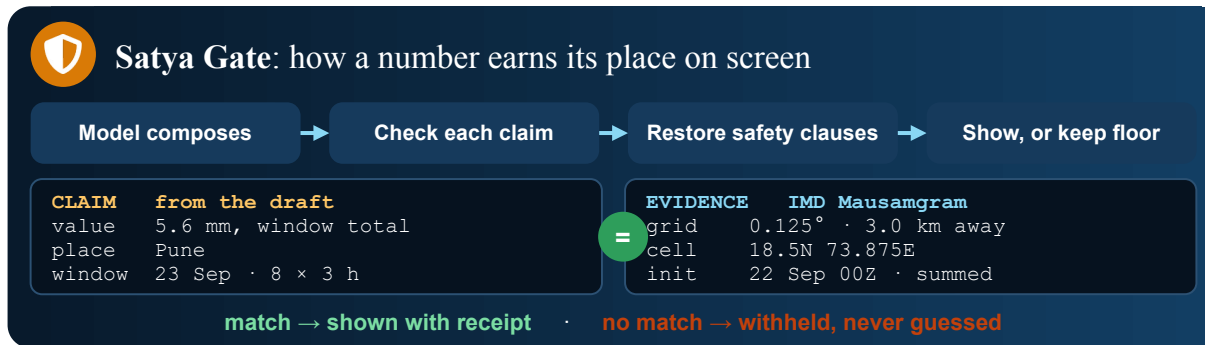
2

TECHNICAL APPROACH

Proposed deployment design · Current core: React, Python, SQLite and governed tools



GOVERNING RULE: tools own factual values; the model plans and explains. Failed checks retain the tool-owned answer.



FEASIBILITY AND VIABILITY

Feasibility analysis · Potential challenges & risks · Strategies to overcome them

Feasibility

273/306

atlas outcomes passed
22 Sep; not forecast skill

- Available components: IMD public feeds, GFS, configured models and Sarvam services.
- 23 Sep regression record: 1,637 + 596 checks; 21-step gate, 0 failed.

Viability

TARGET

India-hosted deployment
proposed, not current

- Proposed sovereign deployment: open models on India-region compute, subject to approval.
- Potential integration: IMD website, UMANG and EOC portals; no deployment claimed.

Practical Implementation

90 days

to a Gujarat pilot:
every district, gu / hi / en

- Scalable: 1 state → every State / UT; 4.3 crore farmers already get IMD agromet SMS.
- Low-end phones and 2G; day-one owner **IMD Met Centre + State EOC**.

Potential Challenges & Risks

- 01 AI hallucination:** an LLM may invent a number, or even a warning.
- 02 Data access:** IMD's APIs need credentials and approvals.
- 03 Connectivity:** patchy rural networks and feature phones.
- 04 Language fidelity:** translation can distort values and negations.
- 05 Foreign AI dependence:** outages, price shifts, questions sent abroad.

Satya Gate checks source-bound claims; a rejected draft retains tool-owned wording.

Public routes to IMD's own data: Mausamgram, 764-district nowcast, 2,071 stations; API adapters switch on with a key.

Planned offline + SMS / IVR: last-answer cache, tiny PWA payloads, voice call-back.

Value-safe translation: Sarvam-Translate output re-checked for numbers, units & dates, else it stays in source.

Proposed sovereign stack: India-hosted models. Configured hosted providers receive context now.

Strategies for Overcoming Challenges

PROPOSED Deployment

NATIONAL CORE
IMD / NIC MeghRaj: models, data, audit logs

STATE NODE
State Data Centre + SEOC: languages, alert relay

DISTRICT EDGE
offline node at the DEOC: answers when links fail

CITIZEN DEVICE
PWA cache · SMS · IVR, no data plan needed

ILLUSTRATIVE COST
≈ ₹26 lakh / yr

Biggest line: 2 × H100 inference on IndiaAI compute at ₹150 / GPU-hr, 24 × 7. No per-question foreign API fees.

**DESIGN GOAL
in India**

DPDP Act 2023
district-level location only

CERT-In 2022
logs kept 180 days in India

MeitY cloud
empanelled, India region

Open standards
CAP 1.2 · WIS 2.0 · WFS

NOW
IMD-first answers · installable PWA · streaming

BEFORE FINALE
Live push on a phone · WIS 2.0 lane · WRF

36-HOUR FINALE
SMS / IVR · cyclone-bulletin OCR · IMD API adapters

90-DAY PILOT
Gujarat · all districts · 3 languages

FUTURE SCOPE
WhatsApp bot · panchayat-level forecasts · AI nowcasting · impact-based warnings



IMPACT AND BENEFITS

Intended benefits and illustrative scenarios · Not measured prototype impact

99 %

of days in Jan–Sep 2025 saw extreme weather in India
Source: CSE 2025 ›

4,064

lives lost in those nine months, up 48 % in four years
Source: CSE 2025 ›

9.47 M ha

of crops hit, four times the 2022 figure
Source: CSE 2025 ›

₹12,500

yearly gain per rain-fed farm household using advisories
Source: NCAER 2020 ›

PS OUTCOMES

✓

Faster dissemination of weather information

✓

Improved public accessibility to forecasts

✓

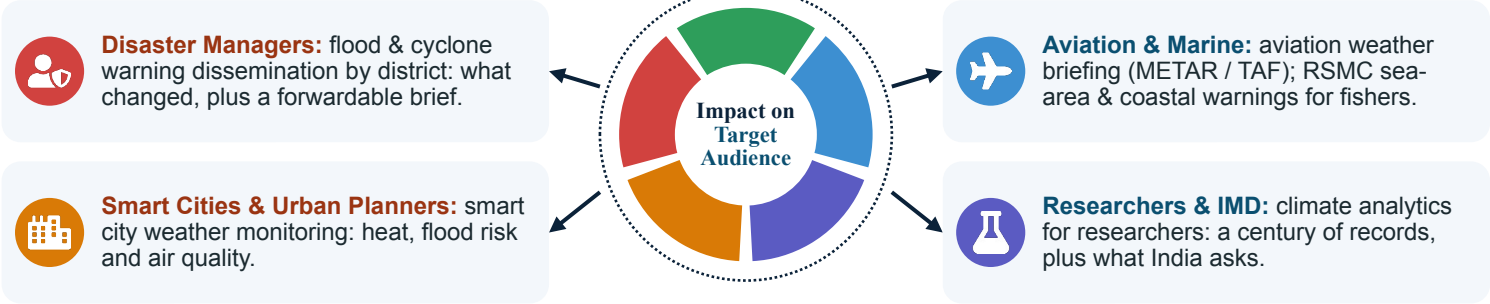
Better disaster preparedness & response

✓

Decision support for agriculture, aviation, marine & urban planning



Farmers: crop-weather advisories at the right hour to spray, sow and irrigate, spoken in their language.



RESEARCH AND REFERENCES

Details and links of the reference and research work

✓ PS decoded
MoES · IMD needs

✓ 70 sources
live-probed

✓ 5 systems
benchmarked

✓ Architecture
designed

✓ Prototype
306 turns run

▶ Now building
live push · WIS 2.0

Capability comparison

✓ yes · ~ partial · × no

	IMD apps	Weather apps	AI chatbots	WeatherGPT
Data stays in India	✓	×	×	~
Ask in natural language	×	×	✓	✓
Official IMD warnings	✓	×	×	✓
No invented numbers	✓	✓	×	~
Source shown on every answer	~	×	×	~
22 languages + voice	~	~	~	~
SMS / IVR reach	~	×	×	~
Forecast · advice · climate in one	×	×	×	✓
IMD + global models, compared	×	×	×	✓

IMD apps studied: [Mausam app](#) · [Meghdoot](#) · [Mausamgram](#)

Official Data, Standards & Research Sources

- **SIH26068**: Ministry of Earth Sciences · India Meteorological Department. [sih.gov.in/sih2026PS](https://gov.in/sih2026PS)
- **IMD Mausam**: forecasts, warnings, bulletins. mausam.imd.gov.in
- **IMD GeoServer**: district-warning layers. reactjs.imd.gov.in/geoserver
- **Mission Mausam**: Cabinet note, 2024. pib.gov.in
- **WMO WIS 2.0 · OASIS CAP 1.2**: community.wmo.int
- **NDMA SACHET**: national CAP alert portal. sachet.ndma.gov.in
- **IMD Pune open data**: rainfall & temperature since 1901. dsp.imdpune.gov.in
- **data.gov.in · CERT-In**: open IMD datasets · log directions. data.gov.in

- **NOAA GFS · Open-Meteo**: numerical weather prediction. open-meteo.com
- **Lewis et al., 2020**: Retrieval-Augmented Generation. [arXiv:2005.11401](https://arxiv.org/abs/2005.11401)
- **Ji et al., 2023**: hallucination in NLG, ACM CSUR. [arXiv:2202.03629](https://arxiv.org/abs/2202.03629)
- **NCAER, 2020**: value of weather advisories. [NCAER2020.pdf](https://ncaer2020.pdf)
- **CSE · Down To Earth**: Climate India 2025. cseindia.org
- **Sarvam AI**: sovereign LLM, IndiaAI Mission; Saaras, Bulbul, Translate. sarvam.ai/models
- **IndiaAI compute**: sovereign GPUs for Indian models. indiaai.gov.in

Field & Applied Research

- **70 official sources** live-probed; **public routes to IMD's own products** found (Mausamgram, nowcast, stations).
- **306-turn atlas**: 273 passed outcomes (22 Sep), including honest refusals; not forecast accuracy.
- **8 multi-turn probes**: 7 held end to end.
- Aligned with **Mission Mausam**, the **IndiaAI Mission**, **Bhashini** and the UN **Early Warnings for All**.

WHAT THE RESEARCH CHANGED

Hallucination surveys → **Satya Gate**
RAG literature → **cited bulletin answers**
NCAER advisory value → **Krishni Salah**
CAP 1.2 · WIS 2.0 → **Chetavani Relay**

Watch the demo · 7:15



Scan or click the QR code

youtu.be/NVT6EnkIEVQ

Open GitHub: code, README and evidence

IMD's evaluation parameters → our answer

Accuracy & relevance

Claim checks + receipts; forecast skill unvalidated

Response latency

Streaming UI; cold retrieval can take longer

Multilingual capability

Hindi / Gujarati demos; broader acceptance open

UI & accessibility

Responsive web; real-device acceptance open

Scalability & innovation

Local SQLite; production scale untested

Real-time integration

IMD warnings, nowcast, Mausamgram connected

Voice for rural access

Hindi ASR demo; spoken-answer UI incomplete

